

AJAY GOWDA

Phoenix, AZ, USA

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PROFESSIONAL SUMMARY

Data-driven professional with 10+ years of experience in **technology innovation, data architecture engineering, and cross-functional product deployments**. Skilled in leading cross-functional teams to develop scalable data pipelines, automate testing, and deliver actionable insights that drive millions in revenue and optimize capital expenditure. Strong background in executive **stakeholder management, mentoring team** and translating complex business challenges to scaled data solutions.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, HTML, Structured Text (PLC), XML, C/C++, MATLAB/Simulink

Technologies: AWS, Linux, Gemini CLI, GitHub Copilot SDK, Model Context Protocol (MCP), Docker, GIT

WORK EXPERIENCE

REPUBLIC SERVICES

Phoenix, USA

Sr. Manager, Operations Technology

Sept 2023 – Present

- Established, and currently lead a high-performing, cross-functional team of Analysts, Engineers and Managers (spanning IT, Economics, and Automotive backgrounds) to drive enterprise-wide adoption of new fleet technology.
- Building an **OWL-based ontology layer** that enables AI agents to execute context-rich, cross-database queries across the company's data platforms.
- Directed the **rapid development** of an AI camera-based safety coaching platform, scaling from prototype to full deployment across the frontline fleet within a year, resulting in an over 80% reduction in safety events.
- Led continuous improvement of billing and analytics for an AI refuse contamination system processing \$200M+ annually; implemented **automated KPI reporting and variance analysis** to identify degradation in technology, people, or processes.
- Engineered a semi-automated, **complex ETL data pipeline** and billing process for AI contamination system by aligning Revenue Management and Field Operations, accelerating time-to-value by 6 months generating \$40M in revenue.

Manager, Technical Operations

Jun 2022 – Aug 2023

- Designed and developed comprehensive EV viability models incorporating route data (miles, lifts), geographic terrain, and custom energy models, directly **guiding \$55M+ in capital expenditure (CapEx)** across 200+ vehicles and 30 locations.
- Partnered with Oshkosh Corp to engineer the first fully integrated electric refuse vehicle; established customer requirements for improved cab and safety specifications, and successfully piloted the prototype via targeted field training.

EVBOX

Amsterdam, NL

Systems Validation Engineer

Sep 2020 – May 2022

- Architected and deployed an **AWS-based centralized data acquisition infrastructure** (InfluxDB, Grafana) to ingest and visualize all internal hardware validation test data, enabling rapid analysis and cross-functional decision-making.
- Engineered automated firmware regression testing suites comprising 64 targeted tests, eliminating 2 days of manual execution and report writing per release cycle.
- Designed and built a scalable Hardware-in-the-Loop (HIL) test bench to evaluate charger performance under extreme temperature profiles, delivering critical data that informed product design and significantly reduced production bug rates.

ETERNAL SUN SPIRE

The Hague, NL

Development and Validation Engineer

Aug 2018 – Jul 2020

- Designed and developed control architectures for complex Photovoltaic Test Equipment, ensuring high data accuracy and compliance with AAA testing standards for optimal performance and data acquisition.
- Conducted root-cause analysis and troubleshooting for prototype and production equipment across lab and manufacturing environments, successfully reducing operational downtime by 20%.

GENERAL MOTORS

Milford, USA

Hybrid Powertrain Development Engineer

Jan 2016 – Aug 2018

- Captured and analyzed millisecond-level telemetry data across all powertrain Electronic Control Units (ECUs) during field tests (test tracks, extreme locations) and advanced test cells to evaluate system behavior.

Algorithm Development Engineer – Thermal Controls

Jul 2015 – Jan 2016

- Developed and rigorously tested thermal control algorithms for liquid-cooled high-voltage battery systems; proactively identified and resolved software anomalies prior to consumer release to ensure product safety and reliability.

EDUCATION

UNIVERSITY OF WASHINGTON

Seattle, USA

Master of Science in ME, Control Systems

Sep 2013 – May 2015